CO5 AT-1

NAME:- R. JANANI SRI
Reg no:- 192511093
Course :- Data Structures
Code :- CSA0307

Pseudocode writing

1. Shortest path with "Toll free" voucher

Problem:-

A graph contains edges with different weight (tolls). We have one voucher that can make the cost of one edge Zero. Find the minimum cost path from source S to destination D

Pseudocode:-

```
Shortest path (G, S, D):
    for every vertex V:
        dist[V][0] = ∞
        dist[V][1] = ∞
    dist[S][0] = 0
    Create Priority queue PQ
    Insert (0, S, 0) into PQ
    while PQ is not empty:
        (cost , u, used) = Remove
```

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minimum from PQ

for each edge (u, v, w):

new cost = cost + w

if newcost < dist[v]

[used]:

dist[v][used] = newcost

Insert (newcost, v, used) into PQ

if used == 0:

newcost = cost

if newcost < dist[v]

dist[v][1] = newcost

Insert (newcost, v, 1) into PQ

return min (dist[D][0], dist[D][1])

Suppose:-

S - - 5 - - > A - - 10 - - > D

\ 8 - - - B - - - 2 - - - D

Without voucher

$S \to A \to D = 5 + 10 = 15$

$S \to B \to D = 8 + 2 = 10$

Using the voucher on the edge $S \to B$:

$S \to B \to D = 0 + 2 = 2$

Shortest path = 2

Time Complexity : $O((V + E) \log V)$

2.) Detecting "Critical Dependencies"

Problem!-

In a task graph, Vertices represent tasks and directed edges represent dependencies

Topological Sort Pseudocode:-

```
TopologicalSort(G):
    Find indegree of every vertex
    Q = empty queue
    for each vertex v:
        if indegree[v] == 0:
            Insert v into Q
    Order = empty list
    while Q is not empty:
        u = Remove from Q
        Add u to Order
        for each neighbour v of u:
            indegree[v]--
            if indegree[v] == 0:
                Insert v into Q

return Order.
```

Example

$A \to B$

$A \to C$

$B \to C$

$C \to D$

Possible topologies

A, B, C, D